Linear Statistical Models

B. Statistical Data Science 2nd Year Indian Statistical Institute

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Exercise Series 4

Exercise 1 (Graded). In this exercise, we fill some gaps (and also redo some steps) in the derivation of the result that the F-test statistic can be obtained by the full-reduced model technique. Recall that our model is $\mathbf{y} = \mathrm{X}\boldsymbol{\theta} + \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathsf{N}_n(\mathbf{0}, \sigma^2\mathrm{I})$. We want to test $\mathcal{H}_0 : \mathrm{A}\boldsymbol{\theta} = \mathbf{0}$ vs. $\mathcal{H}_a : \mathrm{A}\boldsymbol{\theta} \neq \mathbf{0}$. Here, A is a $q \times p$ matrix with $q \leq d$, so that $\mathrm{A}\boldsymbol{\theta}$ is a vector of size q. Moreover, we assume that $\mathrm{rank}(\mathrm{A}) = q$, which means there is no *redundancy* among the components of $\mathrm{A}\boldsymbol{\theta}$.

- The F-test statistic for testing $\mathcal{H}_0$ vs. $\mathcal{H}_a$ under the above setup is

$$F = \frac{(\mathrm{A}\widehat{\boldsymbol{\theta}})^\top \big(\mathrm{A}(\mathrm{X}^\top\mathrm{X})^{-1}\mathrm{A}^\top\big)^{-1}(\mathrm{A}\widehat{\boldsymbol{\theta}})/q}{\mathrm{SSE}/(n-d)}, \qquad \widehat{\boldsymbol{\theta}} = (\mathrm{X}^\top\mathrm{X})^{-1}\mathrm{X}^\top\mathbf{y},\ \mathrm{SSE} = \|\mathbf{y} - \mathrm{X}\widehat{\boldsymbol{\theta}}\|^2.$$

- The full-reduced model technique consists of the following steps:
  - Fit the *full* model: $\mathbf{y} = \mathrm{X}\boldsymbol{\theta} + \epsilon$.

    Calculate $\mathrm{SSE} = \min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathrm{X}\boldsymbol{\theta}\|^2$

    $\mathrm{df} = n -$ number of free parameters of the full model $= n - d$.
  - Fit the *reduced* model (under $\mathcal{H}_0$): $\mathbf{y} = \mathrm{X}\boldsymbol{\theta} + \epsilon$ subject to $\mathrm{A}\boldsymbol{\theta} = \mathbf{0}$.

    Calculate $\mathrm{SSE}_{\mathcal{H}_0} = \min_{\boldsymbol{\theta}:\mathrm{A}\boldsymbol{\theta}=\mathbf{0}} \|\mathbf{y} - \mathrm{X}\boldsymbol{\theta}\|^2$

    $\mathrm{df}_{\mathcal{H}_0} = n -$ number of free parameters of the reduced model $= n - (d - q)$.

    Note: Since we are adding q many linear restrictions on d parameters, the number of *free* parameters is reduced by q. A more rigorous way to see this is the following. Since A is full row-rank, the rows of A are linearly independent. So, we can extend it into a $d \times d$ non-singular matrix $\widetilde{\mathrm{A}}$ by adding linearly independent rows. If we define $\widetilde{\boldsymbol{\theta}} = \widetilde{\mathrm{A}}\boldsymbol{\theta}$, then $\widetilde{\boldsymbol{\theta}}$ is a *full-rank reparameterization* of $\boldsymbol{\theta}$: It is possible to go from one to the other without loosing anything. Also, fitting the model $\mathbf{y} = \mathrm{X}\boldsymbol{\theta} + \epsilon$ is same as fitting $\mathbf{y} = \widetilde{\mathrm{X}}\widetilde{\boldsymbol{\theta}} + \epsilon$, where $\widetilde{\mathrm{X}} = \mathrm{X}\widetilde{\mathrm{A}}^{-1}$. Now, $\mathcal{H}_0 : \mathrm{A}\boldsymbol{\theta} = \mathbf{0}$ means that the first q elements of $\widetilde{\boldsymbol{\theta}}$ are 0, so that the number of free parameters in $\widetilde{\boldsymbol{\theta}}$ is $d - q$ (under $\mathcal{H}_0$).
  - Calculate $\mathrm{SS}(\mathcal{H}_0) = \mathrm{SSE}_{\mathcal{H}_0} - \mathrm{SSE}$.

    $\mathrm{df}(\mathcal{H}_0) = \mathrm{df}_{\mathcal{H}_0} - \mathrm{df} = q$.
  - Define $T = \dfrac{\mathrm{SS}(\mathcal{H}_0)/\mathrm{df}(\mathcal{H}_0)}{\mathrm{SSE}/\mathrm{df}}$.
- We want to show that $T = F$. To show this, it is enough to show that

$$\mathrm{SS}(\mathcal{H}_0) = \mathrm{SSE}_{\mathcal{H}_0} - \mathrm{SSE} = \widehat{\boldsymbol{\theta}}^\top\mathrm{A}^\top\big(\mathrm{A}(\mathrm{X}^\top\mathrm{X})^{-1}\mathrm{A}^\top\big)^{-1}\mathrm{A}\widehat{\boldsymbol{\theta}}.$$

We know that

$$\mathrm{SSE} = \|\mathbf{y} - \mathrm{X}\widehat{\boldsymbol{\theta}}\|^2 = \|\mathbf{y} - \mathrm{H}\mathbf{y}\|^2 = \|\mathbf{y}\|^2 - \|\mathrm{H}\mathbf{y}\|^2,$$

where $\mathrm{H} = \mathrm{X}(\mathrm{X}^\top\mathrm{X})^{-1}\mathrm{X}^\top$ is the orthogonal projection matrix onto $\mathcal{C}(\mathrm{X})$. Also, in the class, we saw that

$$\mathrm{SSE}_{\mathcal{H}_0} = \|\mathbf{y} - \mathrm{X}\widehat{\widehat{\boldsymbol{\theta}}}\|^2 = \|\mathbf{y} - (\mathrm{H} - \mathrm{H}_\mathrm{A})\mathbf{y}\|^2,$$

where $\mathrm{H}_\mathrm{A} = \mathrm{X}(\mathrm{X}^\top\mathrm{X})^{-1}\mathrm{A}^\top\big(\mathrm{A}(\mathrm{X}^\top\mathrm{X})^{-1}\mathrm{A}^\top\big)^{-1}\mathrm{A}(\mathrm{X}^\top\mathrm{X})^{-1}\mathrm{X}^\top$.

(a) Verify that H_A is an orthogonal projection matrix.

(b) Check that $\mathrm{H}\mathrm{H}_\mathrm{A} = \mathrm{H}_\mathrm{A}\mathrm{H} = \mathrm{H}_\mathrm{A}$. So, H_A projects onto a subspace of $\mathcal{C}(\mathrm{X})$.

(c) Use part (b) to show that $\mathrm{H} - \mathrm{H}_\mathrm{A}$ is an orthogonal projection matrix.

Hint: *See Revision 4 (Linear Algebra), Exercise 2.*

(d) Conclude that $\mathrm{SSE}_{\mathcal{H}_0} = \|\mathbf{y}\|^2 - \|(\mathrm{H} - \mathrm{H}_\mathrm{A})\mathbf{y}\|^2$. Hence, $\mathrm{SS}(\mathcal{H}_0) = \|\mathrm{H}\mathbf{y}\|^2 - \|(\mathrm{H} - \mathrm{H}_\mathrm{A})\mathbf{y}\|^2$.

(e) Show that $\|\mathrm{H}\mathbf{y}\|^2 = \|\mathrm{H}_\mathrm{A}\mathbf{y}\|^2 + \|(\mathrm{H} - \mathrm{H}_\mathrm{A})\mathbf{y}\|^2$. Therefore, $\mathrm{SS}(\mathcal{H}_0) = \|\mathrm{H}_\mathrm{A}\mathbf{y}\|^2$.

(f) Check that $\|\mathrm{H}_\mathrm{A}\mathbf{y}\|^2 = \mathbf{y}^\top\mathrm{H}_\mathrm{A}\mathbf{y} = \widehat{\boldsymbol{\theta}}^\top\mathrm{A}^\top\big(\mathrm{A}(\mathrm{X}^\top\mathrm{X})^{-1}\mathrm{A}^\top\big)^{-1}\mathrm{A}\widehat{\boldsymbol{\theta}}$.

Hint: H_A *is an orthogonal projection matrix.*

(g) Notice an alternate way of finding the degrees of freedom in a linear model: $\mathrm{df} = n - \mathrm{rank}(\mathrm{H}) = n - \mathrm{trace}(\mathrm{H})$. Use this to show that $\mathrm{df} = n - d$ and $\mathrm{df}_{\mathcal{H}_0} = n - (d - q)$. Hence, $\mathrm{df}(\mathcal{H}_0) = q$.

(h) Use (f) and (g) to conclude that $T = F$, as intended.

(i) Verify that under the normality assumption, the likelihood ratio test for $\mathcal{H}_0$ vs. $\mathcal{H}_a$ is equivalent to rejecting $\mathcal{H}_0$ for large values of T. Hence, conclude that the likelihood ratio test and the F test are equivalent for this testing problem.

Exercise 2. Consider a linear model with intercept: $\mathbf{y} = \alpha\mathbf{1} + \mathrm{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, where $\boldsymbol{\epsilon} \sim \mathsf{N}_n(\mathbf{0}, \sigma^2\mathrm{I})$. We want to test $\mathcal{H}_0 : \boldsymbol{\theta} = \mathbf{0}$ vs. $\mathcal{H}_a : \boldsymbol{\theta} \neq \mathbf{0}$.

(a) Show that the least-squares estimator $\widehat{\boldsymbol{\beta}}_{\mathrm{LSE}}$ of $\boldsymbol{\beta}$ satisfies

$$\widehat{\boldsymbol{\beta}}_{\mathrm{LSE}} \sim \mathsf{N}_p(\boldsymbol{\beta}, \sigma^2\mathrm{V}^{-1}), \qquad \text{where } \mathrm{V} = \mathrm{X}^\top\left(\mathrm{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top\right)\mathrm{X}.$$

Hint: *Exercise Series 2, Exercise 4.*

(b) Use part (a) to construct the F-test statistic for this problem. Show that it follows the non-central F distribution $F_{p,n-p-1}(\delta)$. The non-centrality parameter $\delta = \dfrac{\boldsymbol{\beta}^\top\mathrm{X}^\top\left(\mathrm{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top\right)\mathrm{X}\boldsymbol{\beta}}{\sigma^2} \geq 0$, and equality holds if and only if $\boldsymbol{\beta} = \mathbf{0}$.

(c) Use the full-reduced model approach to show that the F-test statistic for this problem equals

$$\frac{n - p - 1}{p}\frac{R^2}{1 - R^2},$$

where $R^2 = \dfrac{\sum_{i=1}^n(\widehat{y}_i - \bar{y})^2}{\sum_{i=1}^n(y_i - \bar{y})^2} = \dfrac{\|\widehat{\mathbf{y}} - \bar{y}\mathbf{1}\|^2}{\|\mathbf{y} - \bar{y}\mathbf{1}\|^2}$ is the coefficient of determination. Show that the F-test is equivalent to rejecting $\mathcal{H}_0$ for large values of R^2.

Note: To conduct the F-test for $\mathcal{H}_0 : \boldsymbol{\beta} = \mathbf{0}$ vs. $\mathcal{H}_a : \boldsymbol{\beta} \neq \mathbf{0}$ in the above setup, it is enough to know the values of R^2, the number of observations n and the number of covariates p.

(d) To perform the test for $\mathcal{H}_0$ vs. $\mathcal{H}_a$, it can be tempting to do the following:

- Fit the model $\mathbf{y} = \mathrm{X}\boldsymbol{\beta} + \boldsymbol{\eta}$.
- Test for $\boldsymbol{\beta} = \mathbf{0}$ in the above model.

Define the test statistic under the above-mentioned approach. Notice that if $\mathbf{y} = \alpha\mathbf{1} + \mathrm{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ is the "true" model with $\boldsymbol{\epsilon} \sim \mathsf{N}_n(\mathbf{0}, \sigma^2\mathrm{I})$, then $\boldsymbol{\eta} \not\sim \mathsf{N}_n(\mathbf{0}, \sigma^2\mathrm{I})$. Find the distribution of $\boldsymbol{\eta}$. Show that the distribution of the SSE under the above-mentioned strategy involves the unknown parameter α. Hence, the corresponding F-test is not valid unless $\alpha = 0$.

Note: We have just witnessed an effect of *model misspecification* on our inference!

Exercise 3 (Simulation study 2)**.** We now know that the F-test statistic and the test statistic obtained from the full-reduced model technique are the same. In this exercise, we verify this via simulations. Generate a dataset with $p = 10$ and $n = 100$, consider a linear model with intercept $\mathbf{y} = \alpha\mathbf{1} + \mathrm{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ for the data, and the problem of testing $\mathcal{H}_0 : \boldsymbol{\beta} = \mathbf{0}$ vs. $\mathcal{H}_a : \boldsymbol{\beta} \neq 0$.

(a) Compute the F-test statistic directly using the formula. Also find the test statistic using the full-reduced model technique. Are the values the same? Repeat the experiment a few times with different generations of data.

(b) Repeat the exercise for the following testing problems.

  (i) The contribution of x_1 is twice as much as x_2 and thrice as much as x_3.

  (ii) The covariates x_3, x_6, x_8, x_9 do not affect the response.

  (iii) The contributions of all the covariates are the same.

Clearly specify the matrix A in each case. Remember that A must be of full row rank.

Exercise 4. Consider a linear model of the following form

$$\mathbf{y} = \mathrm{X}_1\boldsymbol{\theta}_1 + \mathrm{X}_2\boldsymbol{\theta}_2 + \boldsymbol{\epsilon},$$

where $\mathrm{X} = (\mathrm{X}_1\ \mathrm{X}_2)$ is of full column rank. Let $\widehat{\boldsymbol{\theta}}_1$ and $\widehat{\boldsymbol{\theta}}_2$ be the least-squares estimates of $\boldsymbol{\theta}_1$ and $\boldsymbol{\theta}_2$ in the model above. Consider another model

$$(\mathrm{I} - \mathrm{H}_{\mathrm{X}_1})\mathbf{y} = (\mathrm{I} - \mathrm{H}_{\mathrm{X}_1})\mathrm{X}_2\boldsymbol{\theta}_2 + \boldsymbol{\eta},$$

where $\mathrm{H}_{\mathrm{X}_1}$ is the orthogonal projection onto the column space of X_1. Let $\widehat{\widehat{\boldsymbol{\theta}}}_2$ be the least-squares estimate of $\boldsymbol{\theta}_2$ in this model. Show that $\widehat{\boldsymbol{\theta}}_2$ and $\widehat{\widehat{\boldsymbol{\theta}}}_2$ are the same. Observe what happens when X_1 and X_2 are orthogonal to each other.

Hint: *Use Exercise Series 2, Exercise 4.*

Note: The second model is feasible, we know (can compute!) $(\mathrm{I} - \mathrm{H}_{\mathrm{X}_1})\mathbf{y}$ and $(\mathrm{I} - \mathrm{H}_{\mathrm{X}_1})\mathrm{X}_2$. The result above gives us a way of estimating all the parameters of a linear model without inverting large matrices. We need to compute $(\mathrm{I} - \mathrm{H}_{-j})$ for all j though!